

Standard Secondary Mathematics Curriculum

Subject Code: MATH-10 | Official Academic Curriculum

Verified learning outcomes and required chapter competencies for students.

Chapter 1: Real Numbers & Number Systems

- Euclid division lemma
- Fundamental theorem of arithmetic
- Irrational proofs
- Decimal expansions

Chapter 2: Polynomials & Algebraic Expressions

- Geometrical representation of zeroes
- Relationship between zeroes and coefficients
- Division algorithm

Chapter 3: Linear Equations in Two Variables

- Graphical method of solution
- Substitution method
- Elimination method
- Cross-multiplication

Chapter 4: Quadratic Equations

- Standard form $ax^2+bx+c=0$
- Factorisation method
- Completing the square
- Quadratic formula and discriminant

Chapter 5: Arithmetic Progressions

- nth term of an AP
- Sum of first n terms
- Real-life word problems

Chapter 6: Introduction to Trigonometry

- Trigonometric ratios sin cos tan
- Ratios of specific angles 30 45 60
- Complementary angles
- Trigonometric identities

Chapter 7: Statistics & Probability

- Mean median mode of grouped data
- Cumulative frequency graphs
- Empirical probability and outcomes